

Work Session Documentation: Slave Board Balancing and Temperature Sensing

Subsystem: Slave Board | Equipment: bench supply, DMM, thermocouple, programmable voltage sources

Session Objective

The objective of this work session was to verify that the slave board could perform passive cell balancing and correctly report the hottest thermistor group through the temperature selection circuit. The session also recorded thermal observations from the balancing path because resistor heating was a known design concern.

Raw Data and Observations

Measurement	Raw Observation	Interpretation at Time of Test
Cell input for balancing	Bench supply set to 4.2 V.	Represents full-charge single-cell condition.
Balancing current	Measured discharge current was 22.4 mA.	Meets ≥ 20 mA prototype balancing requirement.
Bleed resistor heating	Resistor area warmed during enabled balancing.	Power is localized; thermal spreading is necessary.
Temperature reference	Thermocouple compared against LTC6811 reported temperature.	Maximum observed error was 1.8 C.
Max-selection test	Three distinct voltage inputs were permuted.	Output followed selected worst-case input for all tested permutations.

What Was Tried

A 4.2 V input was applied to the cell input and balancing was enabled through the LTC6811 control path. Discharge current was measured with a DMM. Separately, the thermistor measurement path was tested by comparing LTC6811 readings against a thermocouple and by applying three different voltages into the analog selection circuit to confirm that the circuit selected the correct worst-case thermal signal.

What Happened

The balancing current was measured at 22.4 mA, which satisfied the requirement. The resistor area showed noticeable heating, consistent with the design expectation that passive balancing converts removed cell energy into heat. The temperature sensing circuit stayed within the ± 4 C requirement, with a maximum measured error of 1.8 C. The selection circuit behaved consistently during input permutations.

Inference

The slave board met the electrical verification goals, but the session reinforced that passive balancing must be treated as both an electrical and thermal design problem. The current level was sufficient for the prototype, but resistor power dissipation and local heating justified the use of higher-rated resistors, thermal vias, copper area, and careful spacing from MOSFETs and sense traces.